CHE 201 Spring 2020 HW 8

Roxanna Mendoza

TOTAL POINTS

100 / 100

QUESTION 1

1 1 **10 / 10**

✓ **- 0 pts** Correct

- 1 pts Wrong a)

QUESTION 2

2 2 **15 / 15**

✓ **- 0 pts** Correct

- 3.5 pts Wrong a) i

- 4 pts Wrong a)ii

- 3.5 pts Wrong b)i

- 4 pts Wrong b)ii

QUESTION 3

3 3 **15 / 15**

✓ **- 0 pts** Correct

- 1 pts simplify the answer

- 5 pts Wrong C

- 5 pts Wrong B

QUESTION 4

4 4 **15 / 15**

✓ **- 0 pts** Correct

- 3 pts Wrong a)

- 3 pts Wrong b)

- 3 pts Wrong c)

- 3 pts Wrong d)

- 3 pts Wrong max efficiency

QUESTION 5

5 5 **15 / 15**

✓ **- 0 pts** Correct

QUESTION 6

6 6 **15 / 15**

✓ **- 0 pts** Correct

- 5 pts Wrong a)

QUESTION 7

7 7 **15 / 15**

✓ **- 0 pts** Correct

- 2.5 pts Wrong a)

- 5 pts Wrong a)

- 2.5 pts Wrong b)

- 5 pts Wrong b)

- 2.5 pts Wrong c)

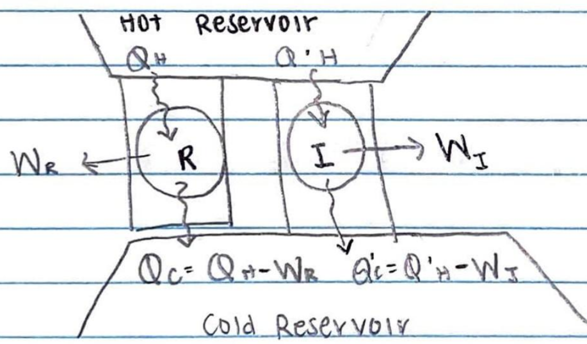
- 5 pts Wrong c)

Homework #8 — Roxanna Mendoza

#1. [change in the amount of entropy contained within the system during some time interval] = [net amount of entropy transferred in across the system boundary during the time interval] − [amount of entropy produced within the system during the time interval]

	Entropy change		Entropy Transfer		Entropy production	
A).	>0	=	0	−	>0	Possible
B).	<0	=	<0	−	>0	Possible
C).	0	=	>0	−	<0	Impossible
D).	>0	=	>0			Indeterminate
E).	0	=	<0	−	>0	Possible
F).	>0	=	<0	−	<0	Impossible
G).	<0	=	<0	−		Indeterminate

#2.



$$\eta_R = \frac{W_R}{Q_H} \qquad \eta_I = \frac{W_I}{Q_H} \qquad \eta_I = \frac{1}{3}\eta_R$$

$$W_R = \eta_R \cdot Q_H \quad \Rightarrow \quad W_I = \frac{Q'_H \cdot \eta_R}{3}$$

$$Q_H = Q'_H$$

$$\frac{1}{3} W_R = W_I \quad \Rightarrow \quad W_R = 3 W_I$$

∴ cycle develops greater network in cycle R.

ii). $Q_C = Q_H - W_R$

$Q'_C = Q'_H - W_I \Rightarrow Q'_C = Q_H - \frac{1}{3} W_R$

∴ Q'_C is greater than Q_C

Which means Irreversible Cycle I would discharge more energy by heat transfer

1 1 **10 / 10**

✓ **- 0 pts** Correct

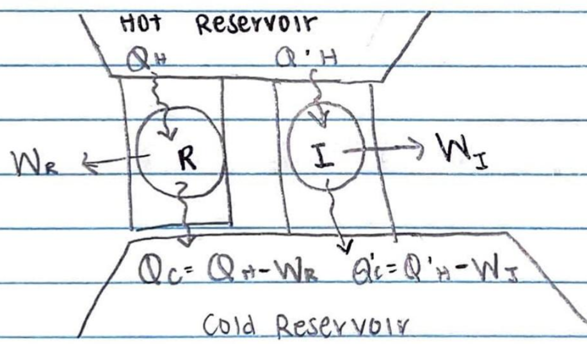
- 1 pts Wrong a)

Homework #8 Roxanna Mendoza

#1. [change in the amount of entropy contained within the system during some time interval] = [net amount of entropy transferred in across the system boundary during the time interval] − [amount of entropy produced within the system during the time interval]

	Entropy change		Entropy Transfer		Entropy production	
A).	>0	=	0	−	>0	Possible
B).	<0	=	<0	−	>0	Possible
C).	0	=	>0	−	<0	Impossible
D).	>0	=	>0			Indeterminate
E).	0	=	<0	−	>0	Possible
F).	>0	=	<0	−	<0	Impossible
G).	<0	=	<0	−		Indeterminate

#2.



$$\eta_R = \frac{W_R}{Q_H} \qquad \eta_I = \frac{W_I}{Q_H} \qquad \eta_I = \frac{1}{3}\eta_R$$

$$W_R = \eta_R \cdot Q_H \quad \Rightarrow \quad W_I = \frac{Q'_H \cdot \eta_R}{3}$$

$$Q_H = Q'_H$$

$$\frac{1}{3} W_R = W_I \quad \Rightarrow \quad W_R = 3 W_I$$

∴ cycle develops greater net work in cycle R.

ii). $Q_C = Q_H - W_R$

$Q'_C = Q'_H - W_I \Rightarrow Q'_C = Q_H - \frac{1}{3} W_R$

∴ Q'_C is greater than Q_C

Which means Irreversible cycle I would discharge more energy by heat transfer

Part B: $W_I = W_R$

i). $\frac{1}{3}\eta_R = \eta_I$

$$\frac{W_R}{Q_H} = \eta_R \quad \& \quad \frac{W_I}{Q'_H} = \eta_I$$

$$\frac{1}{3}\left(\frac{W_R}{Q_H}\right) = \frac{W_I}{Q'_H}$$

$$\frac{1}{3Q_H} = \frac{1}{Q'_H} \Rightarrow 3Q_H = Q'_H$$

The cycle that receives greater energy by heat transfer from hot reservoir is irreversible cycle I.

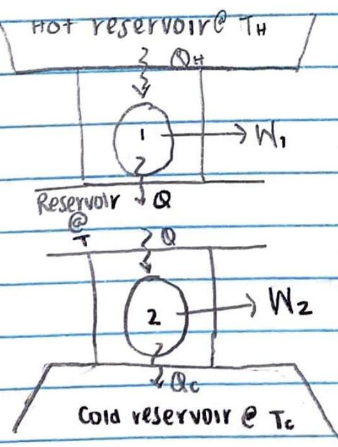
ii). $Q_C = Q_H - W_R$

$Q'_C = Q'_H - W_I \quad \& \quad 3Q_H = Q'_H$

$$Q'_C = 3Q_H - W_I$$

$Q'_C > Q_C$ ∴ irreversible cycle I discharges greater energy by heat transfer.

#3.



2 2 15 / 15

- ✓ **- 0 pts** Correct
- **- 3.5 pts** Wrong a) i
- **- 4 pts** Wrong a)ii
- **- 3.5 pts** Wrong b)i
- **- 4 pts** Wrong b)ii

Part B: $W_I = W_R$

i). $\frac{1}{3}\eta_R = \eta_I$

$\frac{W_R}{Q_H} = \eta_R$ & $\frac{W_I}{Q'_H} = \eta_I$

$\frac{1}{3}\left(\frac{W_R}{Q_H}\right) = \frac{W_I}{Q'_H}$

$\frac{1}{3Q_H} = \frac{1}{Q'_H} \Rightarrow 3Q_H = Q'_H$

The cycle that receives greater energy by heat transfer from hot reservoir is irreversible cycle I.

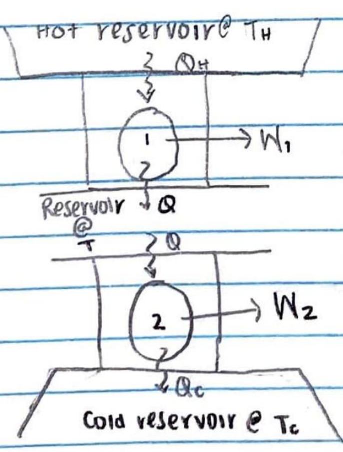
ii). $Q_C = Q_H - W_R$

$Q'_C = Q'_H - W_I$ & $3Q_H = Q'_H$

$Q'_C = 3Q_H - W_I$

$Q'_C > Q_C$ ∴ irreversible cycle I discharges greater energy by heat transfer.

#3.



Part A:

$$\eta_1 = 1 - \frac{Q}{Q_H} \Rightarrow \frac{Q}{Q_H} = 1 - \eta_1$$

$$\eta_2 = 1 - \frac{Q_c}{Q} \Rightarrow \frac{Q_c}{Q} = 1 - \eta_2$$

$$\eta_T = 1 - \frac{Q_c}{Q_H} \cdot \frac{Q}{Q_H}$$

$$\eta_T = 1 - (1-\eta_2)(1-\eta_1)$$

$$= \cancel{1} - (\cancel{1} - \eta_1 - \eta_2 + \eta_2\eta_1)$$

$$\eta_T = \eta_1 + \eta_2 - \eta_2\eta_1$$

Part B:

$$\eta_1 = 1 - \frac{T}{T_H} \quad \& \quad \eta_2 = 1 - \frac{T_c}{T}$$

We know from Part A that $\eta_T = \eta_1 + \eta_2 - \eta_2\eta_1$

$$\eta_T = \left(1 - \frac{T}{T_H}\right) + \left(1 - \frac{T_c}{T}\right) - \left(1 - \frac{T_c}{T}\right)\left(1 - \frac{T}{T_H}\right)$$

$$\eta_T = 1 - \cancel{\frac{T}{T_H}} + \cancel{1} - \cancel{\frac{T_c}{T}} - \cancel{1} + \cancel{\frac{T}{T_H}} + \cancel{\frac{T_c}{T}} - \frac{T_c}{T} \cdot \frac{T}{T_H}$$

$$\eta_T = 1 - \frac{T_c}{T_H}$$

Part C:

$$\eta_1 = \eta_2$$

$$1 - \frac{T}{T_H} = 1 - \frac{T_c}{T}$$

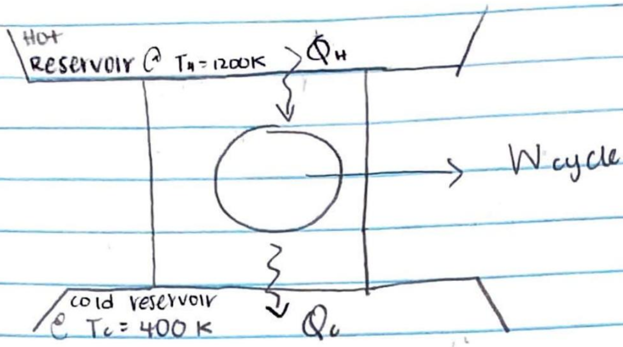
$$\frac{T}{T_H} = \frac{T_c}{T} \Rightarrow \sqrt{T^2} = \sqrt{T_c \cdot T_H}$$

$$T = \sqrt{T_c \cdot T_H}$$

3 3 **15 / 15**

- ✓ **- 0 pts** Correct
- **- 1 pts** simplify the answer
- **- 5 pts** Wrong C
- **- 5 pts** Wrong B

#4.



Part A: $Q_H = 900$ kJ , $W_{cycle} = 450$ kJ

$Q_c = Q_H - W_{cycle} = 900 - 450 = 450$ kJ

$$\oint \frac{dQ}{T} = \frac{Q_H}{T_H} - \frac{Q_c}{T_c} = \frac{900\text{ kJ}}{1200\text{ K}} - \frac{450\text{ kJ}}{400\text{ K}}$$

$$\oint \frac{dQ}{T} = 0.75 - 1.125 = -0.375$$

$\sigma_{cycle} = 0.375 > 0$ Irreversible

Part B: $Q_H = 900$ kJ , $Q_c = 300$ kJ

$W_{cycle} = Q_H - Q_c = 900 - 300 = 600$ kJ

$$\oint \frac{dQ}{T} = \frac{900\text{ kJ}}{1200\text{ K}} - \frac{300\text{ kJ}}{400\text{ K}}$$

$$\oint \frac{dQ}{T} = 0.75 - 0.75 = 0$$ Reversible

Part C: $W_{cycle} = 600$ kJ , $Q_c = 400$ kJ

$Q_H = 600 + 400 = 1{,}000$ kJ

$$\oint \frac{dQ}{T} = \frac{1{,}000\text{ kJ}}{1200\text{ K}} - \frac{400\text{ kJ}}{400\text{ K}}$$

$$\oint \frac{dQ}{T} = 0.8\overline{3} - 1 = -0.1\overline{6}$$ $\sigma_{cycle} = 0.1\overline{6} > 0$ Irreversible

Part D: $\eta = 75\% \Rightarrow 0.75$

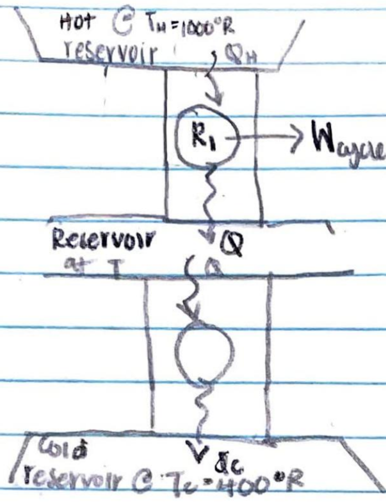
$\eta = 1 - \frac{Q_c}{Q_H} = 0.25 = \frac{Q_c}{Q_H} \Rightarrow Q_c = 0.25 Q_H$

$$\oint \frac{dQ}{T} = Q_H \left(\frac{1}{1200K} - \frac{0.25}{400K} \right) = 0.000208 Q_H < 0$$ Impossible

4 4 15 / 15

- ✓ **- 0 pts** Correct
- **- 3 pts** Wrong a)
- **- 3 pts** Wrong b)
- **- 3 pts** Wrong c)
- **- 3 pts** Wrong d)
- **- 3 pts** Wrong max efficiency

#5.



Part A:

$$W_{R1} = W_{R2}$$

$$Q_H - Q = Q - Q_c$$

$$0 = 2Q - Q_c - Q_H$$

$$Q = \frac{Q_H + Q_c}{2}$$

Reversible

$$\frac{Q}{Q_H} = \frac{T}{T_H}$$

$$Q_H = \frac{T_H}{T} Q$$

Plug into

$$\frac{Q_c}{Q} = \frac{T_c}{T}$$

$$Q_c = \frac{T_c}{T} Q$$

Plug into

$$Q = \frac{\left(\frac{T_H}{T}\right)Q + \left(\frac{T_c}{T}\right)Q}{2} \Rightarrow \cancel{Q} = \cancel{Q}\left(\frac{\frac{T_H}{T} + \frac{T_c}{T}}{2}\right)$$

$$1 = \frac{T_H + T_c}{2T} \Rightarrow T = 0.5(T_H + T_c)$$

$$T = 0.5(1000 + 400) \Rightarrow T = 700\ °R$$

$$\eta_1 = 1 - \frac{T}{T_H} \Rightarrow \eta_1 = 1 - \frac{700}{1000} \Rightarrow \eta_1 = 0.3 = 30\%$$

$$\eta_2 = 1 - \frac{T_c}{T} \Rightarrow \eta_2 = 1 - \frac{400}{700} \Rightarrow \eta_2 = 0.428 = 42.8\%$$

Part B:

$$\eta_{max} = 1 - \frac{T_c}{T_H} \Rightarrow \eta_{max} = 1 - \frac{400}{1000} \Rightarrow \eta_{max} = 0.6 = 60\%$$

$W_{cycle} = W_{R1} + W_{R2}$ since $W_{R1} = W_{R2}$

$$W_{cycle} = 2W_{cycle}$$

5 5 15 / 15

✓ **- 0 pts** Correct

#6. $Q_{OUT} = 120{,}000 \ kJ/h$

$W_{cycle} = 1490 \ kWh$

$$W_{cycle} = Q_{out} - Q_{in} \Rightarrow Q_{in} = Q_{OUT} - W_{cycle}$$

$$Q_{OUT} = \frac{120{,}000 \ kJ}{\cancel{hr}} \left| \frac{24 \ \cancel{h}}{1 \ \cancel{day}} \right| 14 \ \cancel{days} = 40.32 \times 10^6 \ kJ$$

$$W_{cycle} = 1490 \ \cancel{kW}h \left| \frac{1 \ kJ/\cancel{s}}{1 \ \cancel{kW}} \right| \frac{3600 \ \cancel{s}}{1 \ h} = 5.364 \times 10^6 \ kJ$$

$$Q_{in} = (40.32 - 5.364) \times 10^6 \ kJ \Rightarrow Q_{in} = 34.956 \times 10^6 \ kJ$$

Part B: $Cop = \dfrac{Q_{out}}{W_{cycle}} = \dfrac{40.32 \times 10^6 \ kJ}{5.364 \times 10^6 \ kJ} = 7.52$

Part C:

$$T_H = 21 + 273.15 = 294.15 \ K$$

$$T_C = 9 + 273.15 = 282.15 \ K$$

$$C.O.P_{reversible} = \frac{T_H}{T_H - T_C}$$

$$= \frac{294.15 \ K}{294.15 - 282.15 \ K}$$

$$C.O.P_{reversible} = 24.51$$

6 6 **15 / 15**

✓ **- 0 pts** Correct

- 5 pts Wrong a)

#7. At $P = 1\ MPa$

$T_{SAT} = 179.91°C + 273.15 = 453.06\ K$

$h_g = 2778.1\ kJ/kg$

$h_f = 762.79\ kJ/kg$

At $P = 20\ kPa$

$T_{SAT} = 60.06°C + 273.15 = 333.21\ K$

$h_f = 251.38\ kJ/kg$

$h_{fg} = 2358.3\ kJ/kg$

$X_2 = 88\% = 0.88$

$h_2 = h_f + X h_{fg} = (251.38) + (0.88)(2358.3) = 2326.684\ kJ/kg$

$X_3 = 18\% = 0.18$

$h_3 = h_f + X h_{fg} = (251.38) + (0.18)(2358.3) = 675.874\ kJ/kg$

Part A: $\dot{Q}_{in} = h_1 - h_4 = 2015.31\ kJ/kg$

$\dot{Q}_{out} = h_2 - h_3 = 1650.81\ kJ/kg$

$$\oint \frac{dQ}{T} = \frac{2015.31}{452.91} - \frac{1650.81}{333.06} = -0.5068$$

$\sigma_{cycle} = 0.5068$ which is < 0 therefore irreversible

Part B:

$\dot{W}_t = h_1 - h_2 = 451.416\ kJ/kg$

$\dot{W}_p = h_4 - h_3 = 86.916\ kJ/kg$

$W_{net} = \dot{W}_t - \dot{W}_p = 364.5\ kJ/kg$

$$\eta = \frac{W_{net}}{Q_{in}} = \frac{364.5}{2015.31} = 0.1809 \times 100 = 18.09\%$$

Part C:

$$\eta_{carnot} = 1 - \frac{T_{SAT}(20)}{T_{SAT}(1)} = 1 - \frac{333.21}{453.06}$$

$= 0.264 = 26.4\%$

Thermal efficiency is less than Carnot efficiency b/c irreversible process.

7 7 **15 / 15**

- ✓ **- 0 pts** Correct
- **- 2.5 pts** Wrong a)
- **- 5 pts** Wrong a)
- **- 2.5 pts** Wrong b)
- **- 5 pts** Wrong b)
- **- 2.5 pts** Wrong c)
- **- 5 pts** Wrong c)